Candidate's Name.....Signature: .....

545/
1
CHEMISTRY
Paper 1
202
6
2
hours

CHEMISTRY DEPARTMENT EXAMINATIONS BOARD

Uganda Certificate of Education

CHEMISTRY

Paper
1

2
hours

INSTRUCTIONS TO CANDIDATES:

This paper has **six** *examination items which are divided into* **two** *sections*; **A** *and* **B**.

Section **A** *has* **two compulsory** *items. Answers to section* **A must** *be written in the spaces provided.*

Section **B** *has* **two parts; I** *and* **II**. *Answer* **one** *item from* **each** *part. Answers to section* **B must** *be written in the answer booklet(s) provided.*

Answer **four** *items in all.*

Any additional item answered will **not** *be scored.*

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1	2	3	4	5	6

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Turn Over

SECTION A

Answer **all** *items from this section in the spaces provided.*

Item 1

One morning , the children found a panga left in the garden covered with reddish brown bcoat . The children asked the ir father to explain to them what happened to the panga

Task

As a learner of chemistry, write a massage the father delivered to the children and in your massage, include the:

(a) Discover the name of the brown coating on the panga:

_____

_____

_____

ii). the factors that caused the formation of the brown coating on the panga

_____

_____

_____

_____

(b) the properties any one factor that caused the brown coating on the panga.

_____

_____

c). the uses of the factor above in daily life

(c) Realize the side effects of the brown coating on the materials and its mitigations

Item 2

An industry wants to produce lime for treating the acidic soils . The production of the lime requires heating of lime stone strongly according to the equation below

$$CaCO_3\,(s) \longrightarrow CaO\,(s) + CO_2\,(g)$$

The industry is also interested in what mass of lime can be produced such that it can make a clear order from the customers but also the gas to be collected since it is use full. Before the production process, an experiment was carried out using 750g of lime stone were strongly heated until there is no further change

Task:
Use your chemistry knowledge to;

(a) Explain

(i) The categories of the products

(*03 scores*)

(ii) The properties of the gaseous product that makes it use full in daily life (*04 scores*)

(b) Calculate the mass of lime produced during the experiment (Ca=40,O=16, C=12). (*04 scores*)

exp

d). Realize the side effects of any one of the above products to the enviroment and its mitigations

SECTION B
PART 1

Answer any one item from this part on the answer booklets provided.

Item3

In Uganda there is shotage of iron which would be used to make many construction materials which is at high demand in the country due to its essential uses like making iron sheet. The locals make it on small scale and it cannot meet the demands but however the government is planning to set up on industry to make good quality iron on large scales with minimal environmental impacts.

However the community members would like to know the process of production of iron and it impacts to the environment.

Task

As a chemistry student, make a write up you will present to the community members upon invitation.

Item 4

The large scale use of sulphuric acid is to manufacture paint , there was high demand for paint in most of the districts of Uganda which attracted the government to hire an investor to set up the sulphuric acid production plant at Namanve but the residents seem not to understand the how the process will be carried out together with its consequences and are resisting the project.

Task

As a chemistry student; write a presentation you can deliver upon invitation.

Answer one item from this part on the answer booklet provided.

Item 5

In Uganda, crude oil has been discovered and as a result , many people have been displaced from their homes with out being compensated and they are resisting the project. This has attracted the attention of NEMA and it is planning to awareness to the residents and you have been invited by the officials to address them.

Task

As a chemistry student, write a presentation you can deliver upon invitation.

Item 6

In Jinja, the quality of water bodies has been lowered due to which have caused a lot of challenges in the city. This has attracted the attention of officials of the city and they are planning to set up a workshop to address this challenge but they have no idea on how to handle this challenge and they have concentrated you for help.

Task

Write a presentation you will deliver upon invitation.

END